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KHWOPA COLLEGE OF ENGINEERING

AFFILIATED TO TRIBHUVAN UNIVERSITY

A Report on

LAB 1:MIDPOINT COMPENSATION OF A UNCOMPENSATED SYMMETRICAL TRANSMISSION LINE

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LAB 1

MIDPOINT COMPENSATION OF A SYMMETRICAL UNCOMPENSATED TRANSMISSION LINE

1 Objectives

To understand the concept of midpoint compensation of an uncompensated transmission line and to simulate the same using MATLAB/Simulink.

2 Software Required

- MATLAB/Simulink.

3 Theory

An uncompensated transmission line is a line where reactive power control is not done. Hence compensation systems are used to add reactive power control in a transmission line. The need for such reactive power control can be understood by studying the voltage profile of an uncompensated transmission line.

To understand , let us consider a short transmission line with only series impedance.

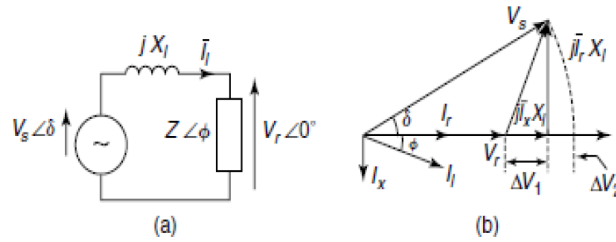


Figure 1: Short Transmission Line Model

From the above figure it is clear that between the sending and the receiving end voltages, magnitude variation as well as a phase difference is created and the most significant part of the voltage drop in the line reactance is due to the reactive component of the load current and to keep the voltages in the load current and to keep the voltage in the line same, we can compensate using two methods: Load compensation and system compensation.

Symmetrical line and Midpoint voltage of Symmetrical Line

A symmetrical line is a transmission line where the sending end and receiving end voltages are equal in magnitude. The power relations of a distributed symmetrical line, we may write,

$$P_R = -P_s = \frac{V^2 \sin \delta}{Z_c \sin(\beta d)} \quad (1)$$

$$Q_r = -Q_s = \frac{V^2 (\cos \delta - \cos(\beta d))}{Z_c \sin(\beta d)} \quad (2)$$

Active and reactive powers are normalized by choosing the SIL as the base.

3.1 Power relations and midpoint conditions in a short symmetrical uncompensated line

Let us consider a simple two machine model where the machines are connected by a purely inductive short transmission line as shown in the figure below:

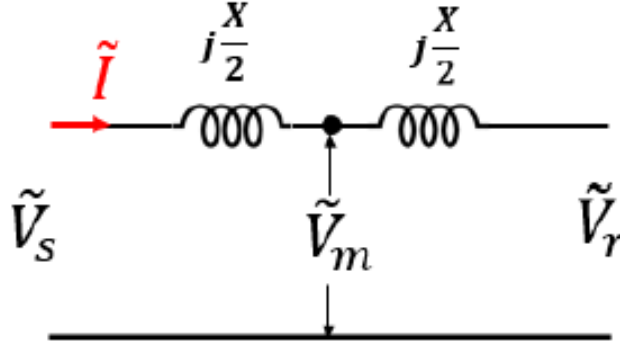


Figure 2: Two Machine Model Connected by a Short Transmission Line

The phasor of the circuit can be realized as :

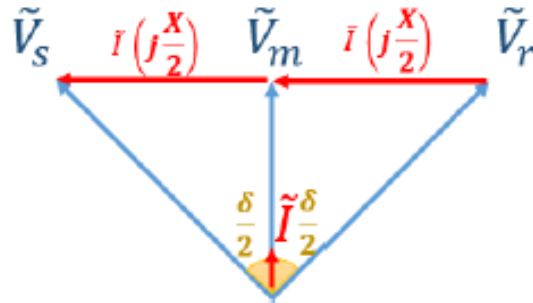


Figure 3: Phasor Diagram of the Two Machine Model

Taking the midpoint voltage as reference, $V_m = V_m < 0^\circ$, we can write the sending end

and receiving end voltages as:

$$V_s = V < \frac{\delta}{2} = V \left(\cos \frac{\delta}{2} + j \sin \frac{\delta}{2} \right) \quad (3)$$

$$V_R = V < -\frac{\delta}{2} = V \left(\cos \frac{\delta}{2} - j \sin \frac{\delta}{2} \right) \quad (4)$$

From the phasor diagram, applying parallelogram law, we can write the midpoint voltage as:

$$2V_m = V_s + V_R \quad (5)$$

$$V_m = V \cos \frac{\delta}{2} < 0^\circ = \frac{V_s + V_r}{2} \quad (6)$$

The current through the line is given by,

$$I = \frac{V_s - V_r}{jX} \quad (7)$$

$$= \frac{2V}{X} \sin \frac{\delta}{2} < 0^\circ \quad (8)$$

For lossless line, the active power at the sending and receiving end are given by,

$$P = V_m I = \frac{V^2}{X} \sin \delta \quad (9)$$

The reactive power is :

$$Q_s = -Q_r = VI \sin(\delta/2) = \frac{V^2}{X} (1 - \cos(\delta)) \quad (10)$$

The voltage profile of a symmetrical line for varying load is as shown in figure:

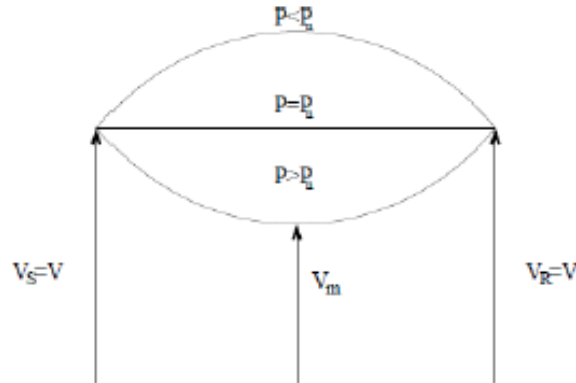


Figure 4: Voltage Profile of a Symmetrical Uncompensated Transmission Line

The power curve after compensation is as shown :

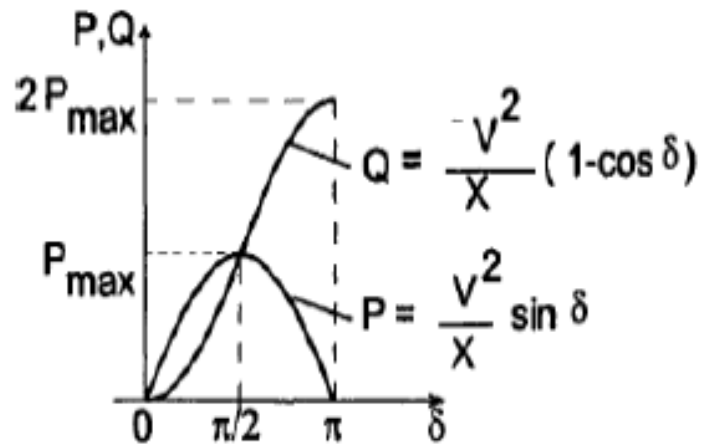


Figure 5: Power Curve after Midpoint Compensation

It is seen that the reactive power compensated is twice the active power transferred.

4 Observation

Uncompensated Transmission Line

The Simulink model of an uncompensated transmission line is as shown below:

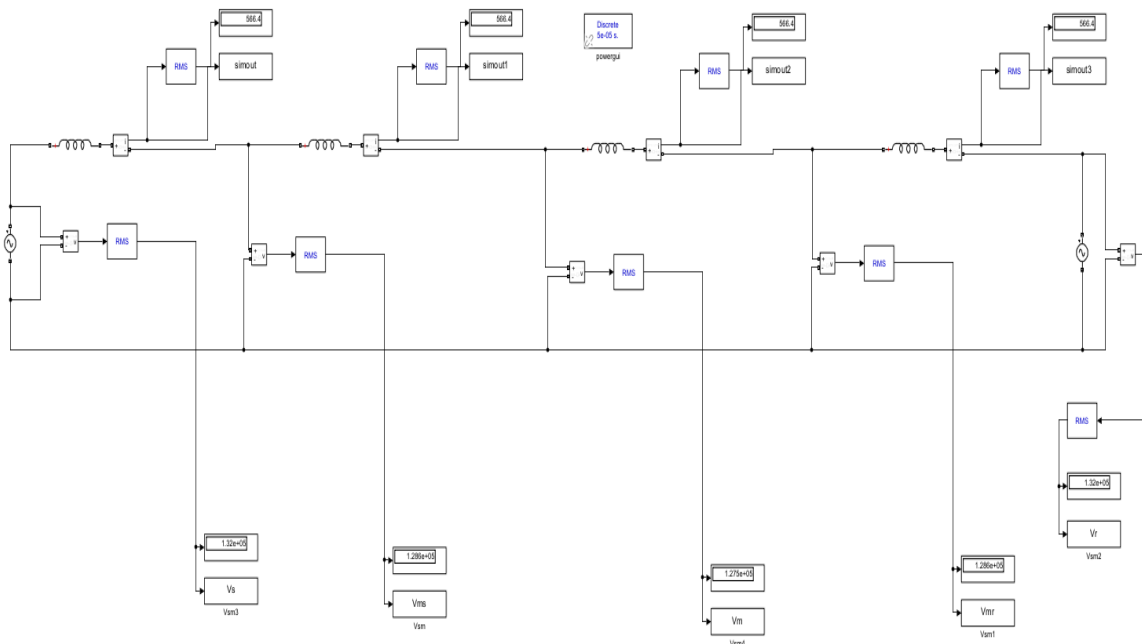


Figure 6: Simulink Model of Uncompensated Transmission Line

The voltage profile of the uncompensated transmission line is as shown below:

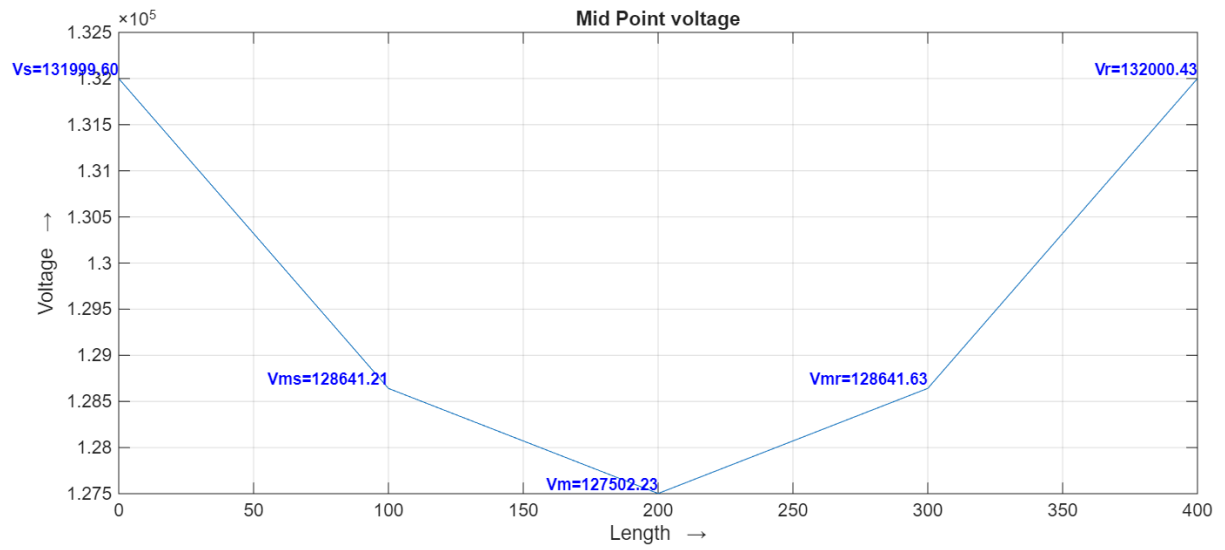


Figure 7: Voltage Profile of Uncompensated Transmission Line

Compensated Transmission Line

The Simulink model of a compensated transmission line is as shown below:

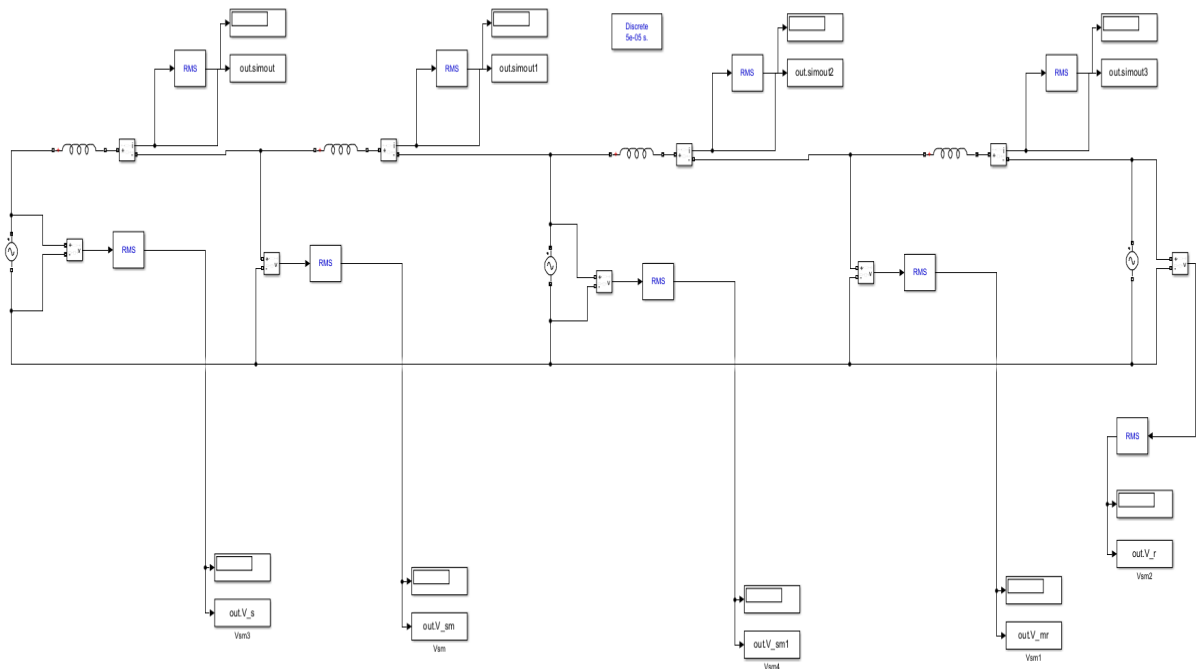


Figure 8: Simulink Model of Compensated Transmission Line

The voltage profile of the compensated transmission line is as shown below:

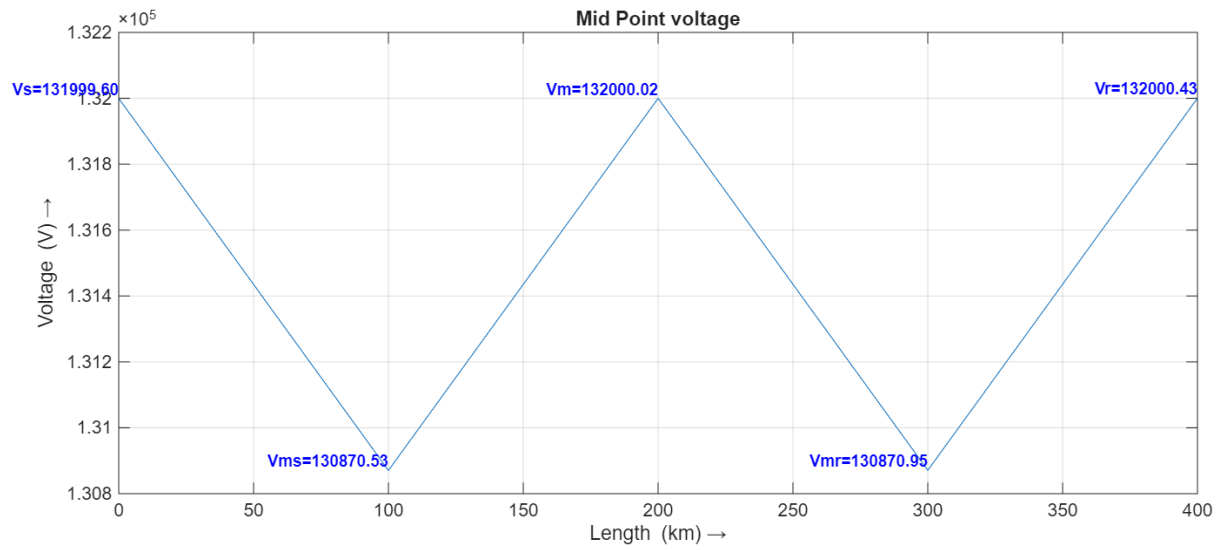


Figure 9: Voltage Profile of Compensated Transmission Line

Compensation of voltage at Quarter Points of the transmission Line

The Simulink model of compensation at quarter points of the transmission line is as shown below:

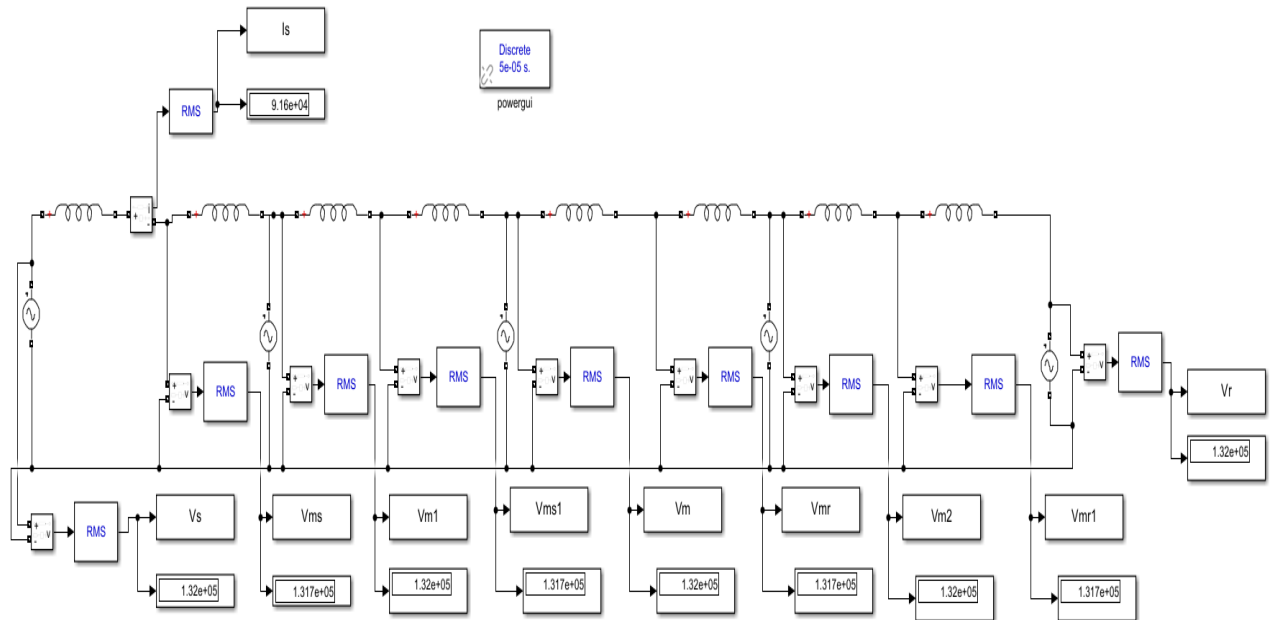


Figure 10: Simulink Model of Compensation at Quarter Points of the Transmission Line

The voltage profile of the compensated transmission line at quarter points($jX/4$) is as shown below:

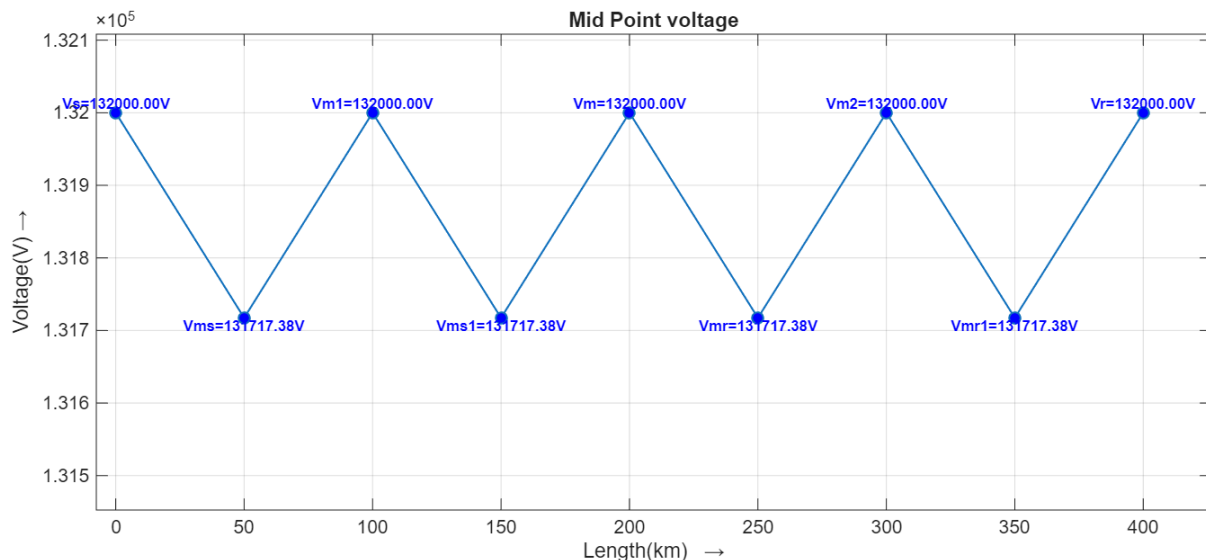


Figure 11: Voltage Profile of Compensated Transmission Line at Quarter Points($jX/4$)

5 Discussion

For the symmetrical uncompensated line, the voltage decreases smoothly from both ends towards the centre, forming a U shaped voltage profile. The midpoint voltage is lower than both the sending and receiving end voltages due to the voltage drop across the line reactance.

With midpoint compensation of $jX/2$, half of the line reactance is cancelled at the centre, so the midpoint voltage rises close to the rated value while two smaller voltage dips appear symmetrically on either side. The voltage profile becomes M-shaped, indicating improved regulation and higher power-transfer capability compared to the uncompensated case.

When the compensation is split into two $jX/4$ sections, reactive support is distributed along the line, producing an almost flat voltage profile with only small ripples. This arrangement gives the best voltage regulation and stability margin, showing that distributed series compensation is more effective than concentrating it at a single point.

6 Conclusion

Hence, we saw that compensating voltages at different points significantly improved the voltage profile of the transmission line.